

Andrew Cheng

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EDUCATION

Stony Brook University

Stony Brook, NY

Bachelor of Science with Honors in Computer Science and Math

Expected Graduation: May 2028

- **GPA: 3.77/4.00** | Relevant Coursework: Computer Vision, Operating Systems, Algorithms

EXPERIENCE

Amazon Web Services

Sep 2026 – Current

Software Engineer Intern (EC2 Hyperplane Networking)

Seattle, WA

- Built a Python CLI to rebuild any VPC's historical network state, validated against replayed production incidents
- Replayed events from restart snapshots across sharded AWS hosts, tracing deletes by ENI ID to rebuild VPC state
- Closed observability gaps in the VPC data-plane controller by logging network ACL and route table state
- Surfaced past VPC state on the team's operations dashboard so on-call engineers can recreate and triage incidents

Roblox

May 2026 – Aug 2026

Software Engineer Intern (Content Core Systems)

San Mateo, CA

- Built a C# AI investigation harness using Microsoft Agents Framework, enabling engineers and cross-functional stakeholders to query asset data in mission-critical production CockroachDB databases using natural language
- Improved on-call efficiency by **16%** by automatically investigating operations and triaging pager duty alerts
- Reduced token usage by **28%** and improved correctness by **17%** on curated evals through tool-use optimizations
- Instrumented per-request logging in a multi-get API serving **10M+** QPS to expose formerly opaque batch failures

Scale AI

May 2025 – Dec 2025

Technical Advisor Intern (Generative AI)

Remote

- Developed ground-truth solutions and verification test suites for **100+** SWE-bench tasks on open-source repos
- Evaluated LLM performance on **1k+** complex reasoning tasks, such as competitive coding and Olympiad math
- Uncovered Gemini deep research failure-modes across **15** domains by testing on **50+** prompt variations

PROJECTS

crowdtest.ai | [Demo Video](#) | *Python, FastAPI, OpenAI Agents SDK, Next.js, TypeScript*

- Created prediction tool to help X creators know how their audience will respond to tweets before they post
- Generated **3.3M** views on a single X post, with the product generating **\$26K** in launch revenue
- Leveraged OpenAI Agents SDK to simulate hundreds of X user reactions in parallel, reducing runtime by **80%**
- Implemented genetic algorithm to refine tweets based on the user's tone, diction, and style in past tweets

Social Stockfish | [Demo Video](#) | *Python, TypeScript, Flask, React, Node.js, Together.ai*

- Won best **Overall Hack** at Columbia Devfest Hackathon; **\$1200 won** and **2M+** views across X and LinkedIn
- Created multi-agent system to optimize messages to likely replies and evaluate conversation trees in context
- Used parallel processing to simulate 5 messages ahead in 243 conversations with Llama 3.3 70B in 10 seconds
- Implemented Monte Carlo simulation to generate probability-weighted optimizations to enhance reasoning
- Created iMessage integration in live chat emulator, showing conversation branches with analysis in real-time

SATA for Segmentation | *Python, PyTorch, HuggingFace Transformers*

- Extended spatial autocorrelation (SATA) module to EoMT segmentation model via runtime layer patching
- Diagnosed and resolved architectural incompatibility improving mIoU from **2.5%** on naive integration to **23.8%**
- Built custom **ADE20K-C** robustness benchmark of **15,000** corrupted images across **15 corruption types** and **5 severity levels** to evaluate segmentation degradation under distribution shift

TECHNICAL SKILLS

Languages: Python, C++, C, C#, Java, JavaScript, TypeScript, SQL, Bash

Machine Learning: PyTorch, TensorFlow, Hugging Face, scikit-learn, OpenCV

Data: PostgreSQL, MongoDB, SQLite, SQLAlchemy, NumPy, Pandas, Matplotlib

Cloud & DevOps: AWS (EC2, CloudWatch), Google Cloud, Docker, GitHub Actions, CI/CD, Linux, Git

Web & Testing: FastAPI, Flask, REST APIs, React, Next.js, Node.js, Tailwind CSS, pytest, Postman

ACTIVITIES

Teamfight Tactics – Ranked 99.9th percentile (Grandmaster **Top 500**) in North America

SBU CompProg – 2x Top **30** at Greater New York ICPC

Chinese Association at Stony Brook (CASB) – Fundraising Chair